Hall Ticket Number:

MR-22

Code:B324CM2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Regular/Supplementary Examinations, December 2024/January 2025

Software Engineering

(Computer Science and Engineering (AI and ML))

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1= 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Write the other name of waterfall model and who invented waterfall model?	1	1	1	1
1. b)	Classify the models in CMMI?	1	4	1	1
1. c)	Distinguish functional requirement?	1	4	1	1
1. d)	Explain requirement validation?	1	2	1	1
1. e)	Discuss analysis and design model?	1	6	2	3
1. f)	Write short notes on architecture patterns?	1	1	2	3
1. g)	Write short notes on validation testing?	1	1	3	3
1. h)	Explain art of debugging?	1	2	3	4
1. i)	Write about quality concepts?	1	1	3	6
1. j)	Simplify software reliability?	1	4	3	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Describe "Software myth"? Discuss on various types of software myths?	5	2	1	1
2. b)	Demonstrate the evolving role of software?	5	2	1	1
	OR				
2. c)	Explain different software applications?	5	2	1	1
2. d)	Describe with the help of the diagram discuss in detail waterfall model?	5	2	1	1
3. a)	Compare functional requirements with non-functional requirements?	5	4	2	1
3. b)	Demonstrate system requirements in a detail manner?	5	2	1	1
	OR				
3. c)	Summarize briefly how requirement validation is done?	5	6	3	3
3. d)	Explain interface specification in detail?	5	2	2	2
4. a)	Explain with a neat diagram of architectural design?	5	2	2	3
4. b)	Analyze mapping data flow into software architecture?	5	4	2	3
	OR				
4. c)	Explain the design concepts in software engineering?	5	2	2	3
4. d)	Illustrate architectural styles and patterns?	5	2	2	4
5. a)	Compare black box testing with white box testing?	5	4	3	5
5. b)	Explain about Metrics for software quality?	5	4	2	1
	OR				
5. c)	Demonstrate art of debugging	5	2	3	4
5. d)	Simplify test strategies for conventional software	5	4	3	5
6. a)	Explain about RMMM Plan?	5	5	3	1
6. b)	Explain about software risks?	5	5	3	2
	OR				
6. c)	Demonstrate risk identification?	5	2	3	2
6. d)	Explain in detail ISO 9000 quality standards?	5	2	2	3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome